

# Build a ServiceNow AI Agent

Incident Investigation & Data Extraction Agent · Claude Code + ServiceNow SDK (Fluent)

An agent that fires when an incident moves to **In Progress**, reads the short description and description, runs targeted web searches via Apify, synthesizes findings, and posts a structured HTML investigation report to the work notes.

## Prerequisites

- A ServiceNow instance with admin access and the AI Agent plugins enabled.
- An Apify account, with an API token from [console.apify.com/account/integrations](https://console.apify.com/account/integrations).
- Node.js installed, plus Claude Code set up (walkthrough: [youtu.be/DUTutpMFGWs](https://youtu.be/DUTutpMFGWs)).
- ServiceNow SDK (@servicenow/sdk) and a Fluent project initialized.

## The prompt (paste into Claude Code)

Create a ServiceNow AI Agent that runs when an incident goes to In Progress. It is an investigation and data extraction agent. Based on the short\_description and description, it does a web search, investigates, finds a solution or recommendation, and pastes that as an HTML report in the comments. Use Apify for the web search via a tool backed by a Flow Designer custom action that calls the Apify actor with a REST Message in the action's script step.

## What gets built

### src/fluent/actions/apify-web-search.now.ts

A Flow Designer custom action (sys\_hub\_action\_type\_definition) named **Apify Web Search**. Input: query (string). Output: search\_results (string, JSON array). A script step (required\_run\_time: instance) uses sn\_ws.RESTMessageV2 to POST to apify-web-search/run-sync-get-dataset-items, reading the token from the system property apify.api.key.

### src/fluent/agents/incident-investigation-agent.now.ts

An AI Agent that triggers when an incident transitions to In Progress (state = 2), wired with three tools:

| Tool                      | Type   | Purpose                                                     |
|---------------------------|--------|-------------------------------------------------------------|
| Lookup Incident           | CRUD   | Fetches short_description, description, and sys_id.         |
| Apify Web Search          | Action | Searches the web via the Apify actor for solutions.         |
| Post Investigation Report | Script | Writes the HTML report to work_notes via GlideRecordSecure. |

Agent instructions: run 2 to 3 targeted searches, synthesize findings, and post a structured HTML report covering root causes, recommended solutions, workarounds, references, and next steps.

## Before deploying

1. Create the system property

2. Authenticate the SDK

3. Deploy

4. Activate the trigger
- Name: apify.api.key Value: your Apify API token.

npx @servicenow/sdk auth --add https://your-instance.service-now.com --type basic

npm run deploy

Deploys as active: false. In AI Agent Studio open your agent, go to Triggers, and activate it.

**Repo:** [github.com/yvain13/SN\\_AI\\_Agents](https://github.com/yvain13/SN_AI_Agents) , star it for more AI agents dropping in this repo.  
Claude Code setup: [youtu.be/DUTutpMFGWs](https://youtu.be/DUTutpMFGWs) · @aibytusharm